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INSTITUTE OF AERONAUTICAL ENGINEERING

NATIONAL AEROMODELLING COMPETITION

Team ID: Aero-250318

Team Name: VIHANG

Aircraft Name: ULUKA

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1 INTRODUCTION

1.1 Background

National Aeromodelling Competition at Techfest, IIT Bombay is a plan to give a common national platform to students whose interests lie on aerospace and other related engineering aspects. It targets empowering players to manufacture, construct and prove fixed-wing aircrafts to specifications. The event focuses on the innovation of payload carrying capability and safe flight missions under real world conditions. It encourages technical competence, teamwork and real life problem-solving through organized qualifier and main rounds. Finally, the competition makes a difference in developing future aerospace talent as it translates what is taught in the classroom to real-life aeromodelling.

1.2 Design Objective

The design mission is to design a lightweight fixed wing RC aircraft and within the size, weight and propulsion limits. It should be capable of achieving stable and safe flight in the absence of automated control support. The aircraft should have a capability of carrying, retaining, and delivering payloads at the needed time. Payload mechanisms and ground operations must be full transmitter controlled. The overall design should be such that the score is maximized by optimization of payload capacity, flight performance and mission efficiency.

1.3 Design Constraints

S. No	Design Feature	Requirement
1	Aircraft Type	Fixed Wing UAV
2	Propulsion System	All Electric
3	Gross Take-Off Weight(estimated)	2412.7 grams
4	Maximum Battery	Max 3 Cells

Table 1 : Design Constraints

1.4 Design Calculations

- The Thrust-to-Weight ratio is **0.935** without payload.
- The centre of gravity of the aircraft without payload is at **261.434mm** from the nose of the aircraft.
- The centre of gravity of the aircraft with payload is at **231.107mm** from the nose of the aircraft and the static margin is **13.466** with payload.

1.5 Airfoil Selection Justification

We have chosen Clark-Y airfoil because it gives high lift and very stable low-speed performance, ideal for small aircraft. The flat bottom surface makes construction easy and ensures straight forward mounting on the wing structure.

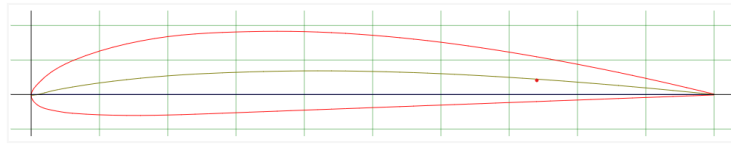


Figure 1 : Clark-Y airfoil

1.6 Overall Configuration

Part	Type
Wing	Rectangle, High Wing
Fuselage	Curved surface at the top and flat surface at the bottom
Tail	Conventional

Table 2 : Overall Configuration

2 POWERPLANT PERFORMANCE

2.1 Motor Selection

Motor	 A2212
KV	1400
Weight (grams)	64
Rated Voltage (LiPo)	3S
Idle Current	0.7A

Table 3 : Motor Selection

2.2 ESC Selection

ESC	 Simonk 30A
Continuous Current(A)	30
Peak Current(A)	40
Weight(grams)	23

Table 4 :ESC Selection

2.3 Battery Selection

BATTERY	 Bonka
Capacity(mAh)	2200
Voltage(V)	11.1
Discharge(C)	35
Weight (grams)	179
Size(mm)	106*34*25

Table 5 : Battery Selection

2.4 Propeller Selection

Propeller	
Size	9.5*4.5
Diameter	9.5in

Pitch	4.5in
Weight	24 grams

Table 6 : Propeller Selection

3 AIRCRAFT PERFORMANCE

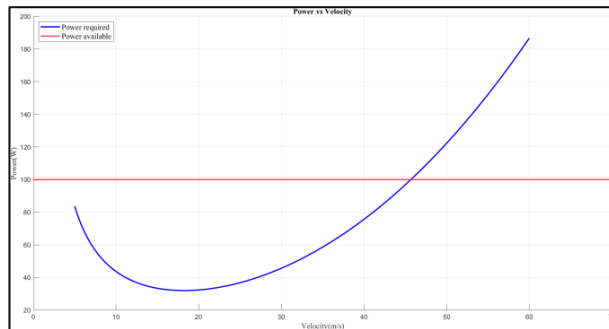


Figure 2 : Power vs Velocity

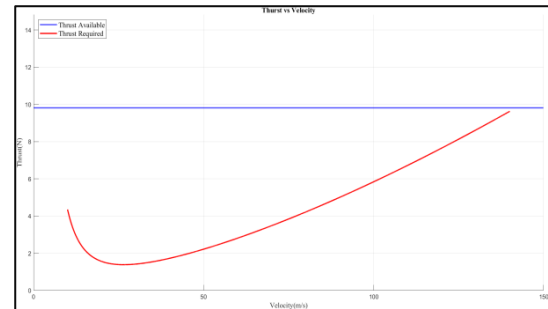


Figure 3 : Thrust vs Velocity

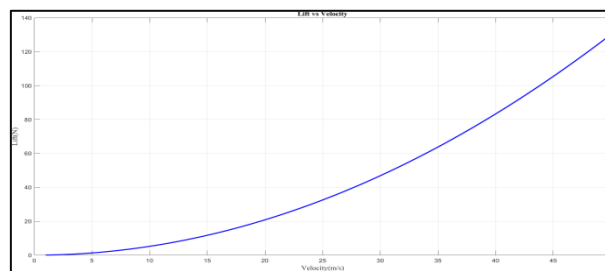


Figure 4 : Lift vs Velocity

4 INITIAL SIZING

4.1 Wing Sizing

The wing is the main part of an airplane that produces most of the lift needed for continuous flight

Parameter	Magnitude(mm)
Span	1160
Surface Area	197200 mm ²
Root Chord	170
Tip Chord	170
Taper Ratio	1

Aspect Ratio	6.82
Wing Loading (with payload estimation)	119.39 N/m ²
Mean Aerodynamic Chord Length	170
Aerodynamic Centre from LE	42.5

Table 7 : Wing Data

4.2 Fuselage Sizing

The fuselage is the main part of the aircraft. It holds the payload, avionics, and support systems.

Fuselage Section	Dimensions(mm)
Total Length of Fuselage	720
Total Length of Fuselage (including) empennage	850

Table 8 : Fuselage data

4.3 Tail Sizing

Horizontal stabilizers adjust for the pitching moments of an aircraft by tilting upward and downward, which helps maintain longitudinal static stability. The vertical tail restricts yawing moments during turns, allowing for lateral movement to the left or right while keeping directional stability.

Parameter	Horizontal Tail	Vertical Tail
Airfoil	NACA 0015	NACA 0015
Span	310 mm	155 mm
Surface Area	34100 mm ²	17050 mm ²
Root Chord	130 mm	130 mm
Tip Chord	90mm	90mm
Taper Ratio	0.69	0.69
Aspect Ratio	2.81	1.5

Table 9 : Tail Data

5 AIRCRAFT WEIGHT ESTIMATION

S. No	Category	Weight(grams)
1	Structural Weight	542.97
2	Avionics Weight	430

3	Estimated Payload Weight	1440
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Table 10 : Aircraft Weight Estimation

Total Weight of the Aircraft without payload is **972.97 grams**

6 CONCEPTUAL DESIGN

6.1 CAD DESIGN

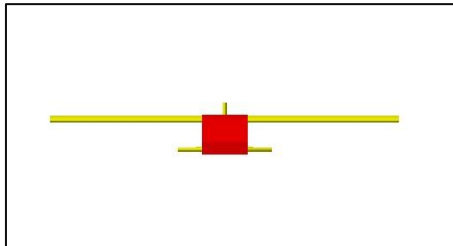


Figure 5 : CAD Model Front View

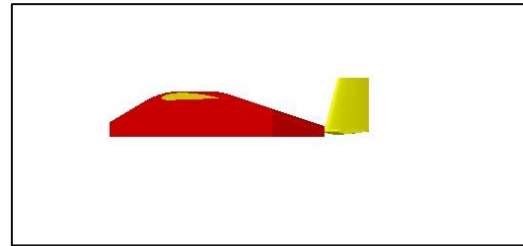


Figure 6 : CAD Model Side View

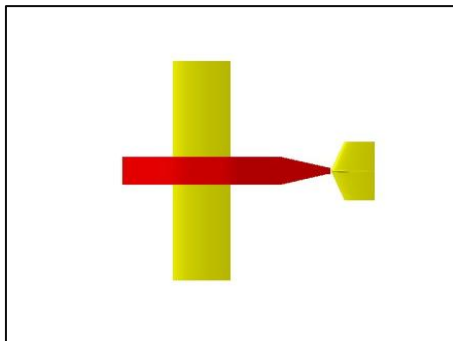


Figure 7 : CAD Model Top View

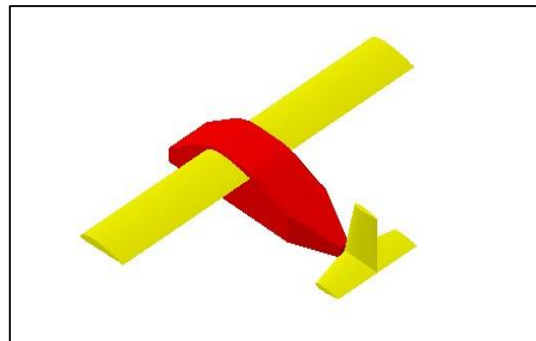


Figure 8 : CAD Model Isometric View

7 PAYLOAD SYSTEM DESIGN

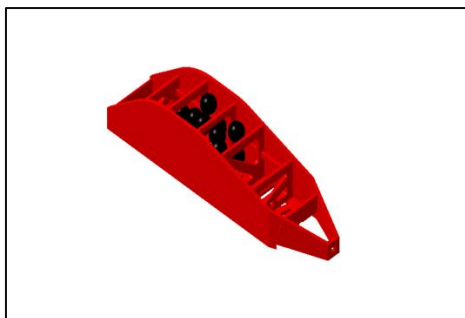


Figure 9 : Payload Compartment

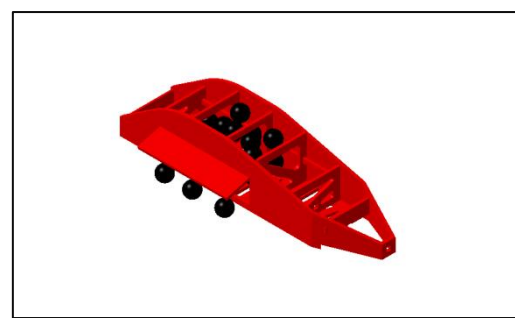


Figure 10 : Payload Dropping Mechanism

We are expecting to carry **32** golf balls placed inside the fuselage **with no separate external payload bay** as we make the structure inside slant where balls are placed. A servo-operated door at the side opens to let the balls roll out. A top opening is provided for quick loading on

the ground. This design minimizes ground time and ensures smooth loading/unloading operations.

8 ANALYSIS

8.1 Structural Analysis of Wing

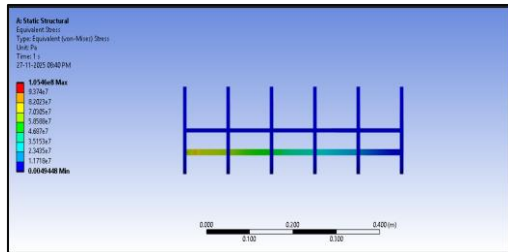


Figure 11 : Equivalent Stress of Wing

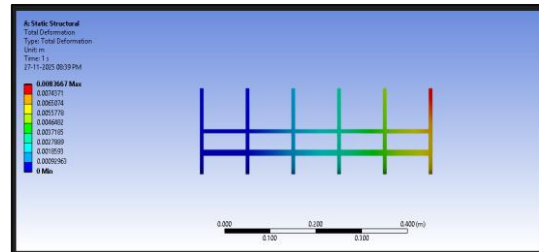


Figure 12 : Total Deformation of Wing

The maximum equivalent stress is **1.0546e8** and maximum deformation is **0.0083667**.

8.2 CFD Analysis of Aircraft

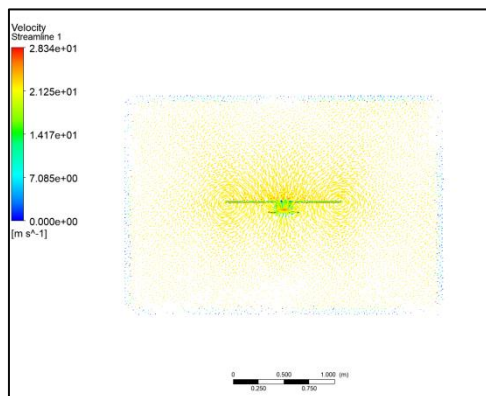


Figure 13 : Velocity Contour of Aircraft

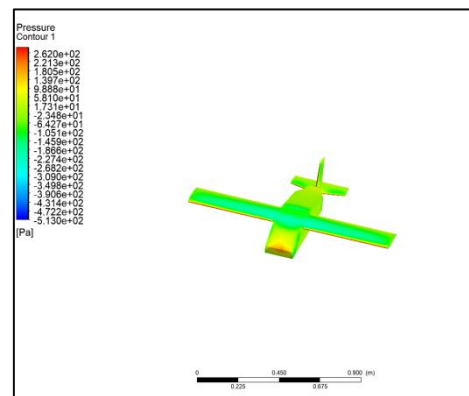


Figure 14 : Pressure Contour of Aircraft

The CFD flow analysis of the aircraft was conducted in the ANSYS FLUENT software in which the aerodynamic characteristics of the aircraft can be studied well. The velocity and pressure contours are shown.

9 COMPETITION STRATEGY

The aircraft's fuselage is a lightweight integrated payload bay constructed from balsa and aluminium rods. The payload area can optimally carry up to thirty-two golf balls. The payload area has a servo-operated door and an inclined ramp so that the golf balls can be quickly released, thereby reducing the time on the ground. The goal in the design of the aircraft's flight characteristics, including low-speed stability; a thirty-second leg mission at a high level of safety; and speed of operations on the ground.

10 STABILITY ANALYSIS

10.1 Longitudinal Static Stability

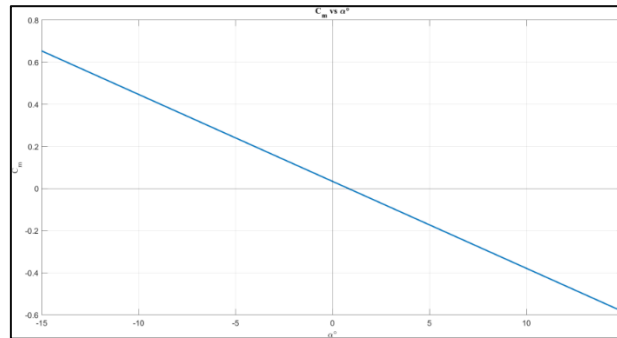


Figure 15 : C_m vs Alpha Graph

The above graph in figure shows the pitching moment behaviour with respect to change in angle of attack (AOA). The pitching moment can be defined as the tendency of the aircraft to pitch nose-up or nose-down due to aerodynamic forces.

10.2 Longitudinal Dynamic Stability

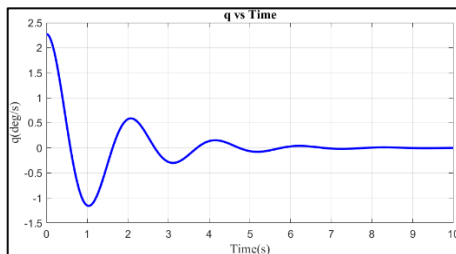


Figure 16 : Pitch Rate(q) vs Time

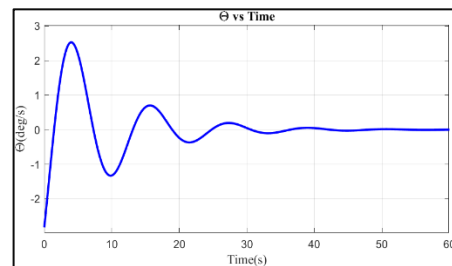


Figure 17 : Pitch Angle (Theta) vs Time

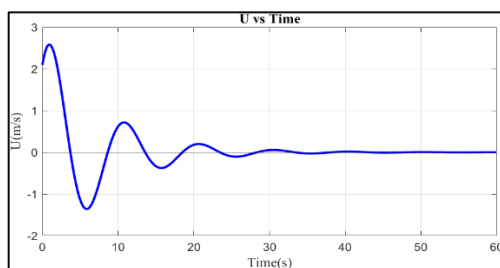


Figure 18 : x- Direction Velocity vs Time

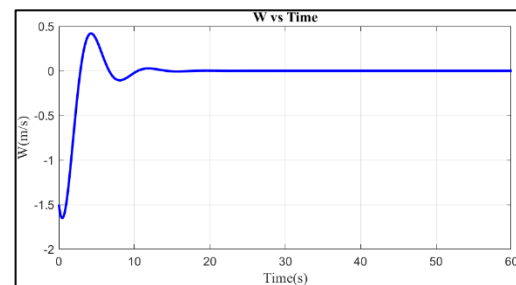


Figure 19 : z- Direction Velocity vs Time

The longitudinal dynamic stability is the behaviour of the aircraft to stabilize back to its equilibrium state after it has been disturbed or perturbed, usually by a range of causes including external loads, turbulence or gusty winds

10.3 Lateral Dynamic Stability

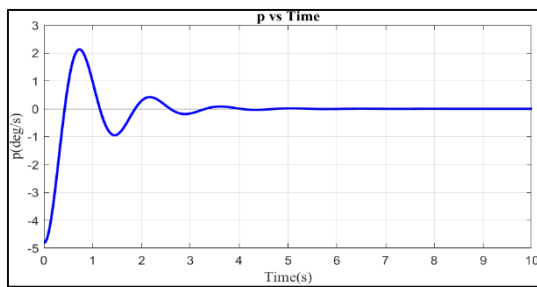


Figure 20 : Roll (p) vs Time

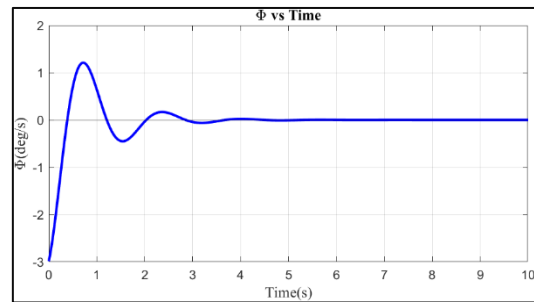


Figure 21 : Bank Angle (phi) vs Time

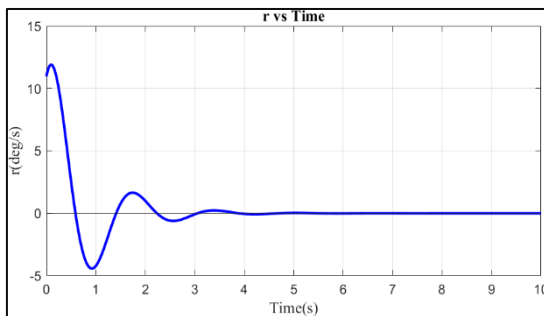


Figure 22 : Yaw (r) vs Time

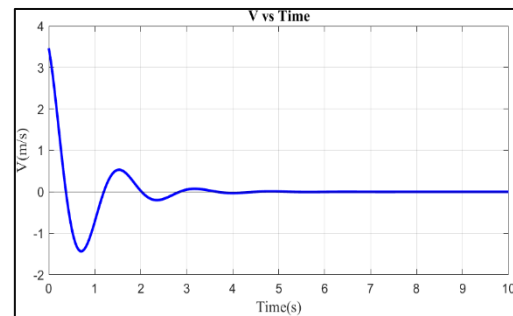


Figure 23 : Velocity vs Time

The response of the aircraft to the gusts in the y-direction has been considered in the graphs. The graphs drawn indicate the lateral stability of the aircraft the parameters are investigated post the disturbance, where the deviation in the plots is damped over time, showing the capability of the aircraft to restore equilibrium.

11 FLIGHT PERFORMANCE

11.1 Flight Phase

S.No.	Phase	Velocity(m/s)
1	Take- Off Velocity	15.35
2	Cruise Velocity	22.15
3	Stall Velocity	12.79

Table 11 : Velocity Data

11.2 Aerodynamic Performance

$$\text{Lift} = 1/2 * \rho * v^2 * S * C_L = 27.121 \text{ N} \quad \text{Drag} = 1/2 * \rho * v^2 * S * C_D = 1.95 \text{ N}$$

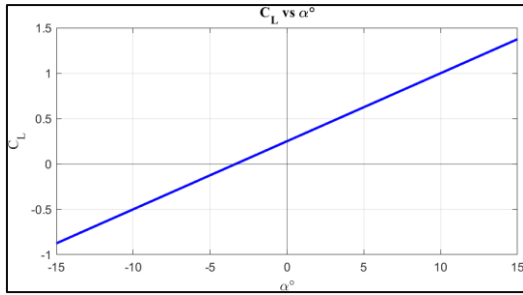


Figure 24 : C_L vs Alpha

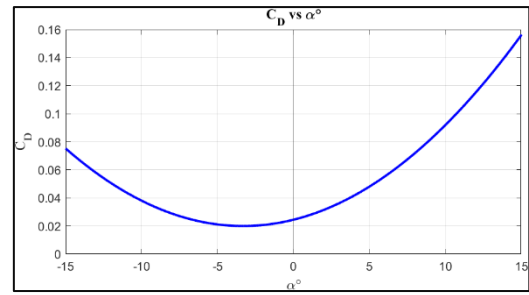


Figure 25 : C_D vs Alpha

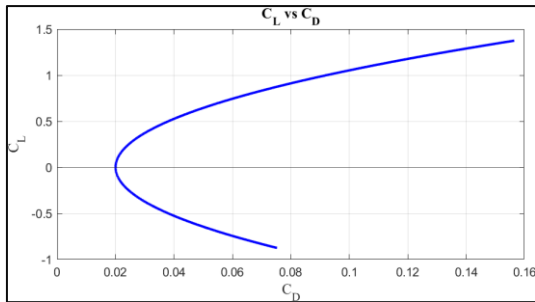


Figure 26 : C_L vs C_D

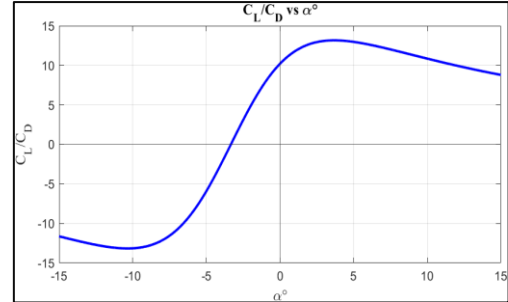


Figure 27 : C_L/C_D vs Alpha

11.3 Velocity Behaviour

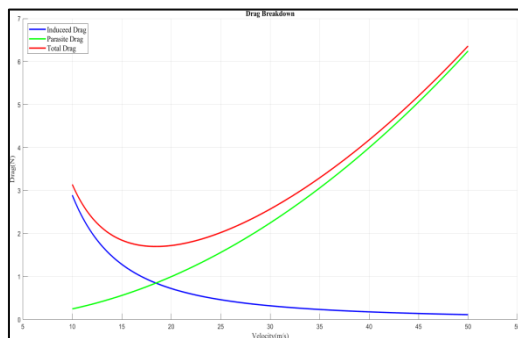


Figure 28 : Drag Breakdown

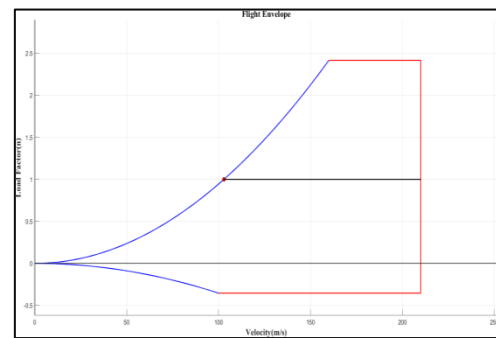


Figure 29 : Flight Envelope

The lift produced by the aircraft at the cruise speed of **22.15 m/s** is about **27.121 N**, which is equal to the amount of lift needed to support its weight for stable, level flight. The amount of drag created is also only **1.95 N**. This cruise point is within the safe flight envelope and provides sufficient control authority and energy reserves for controlling the aircraft during payload missions.

12 FABRICATION

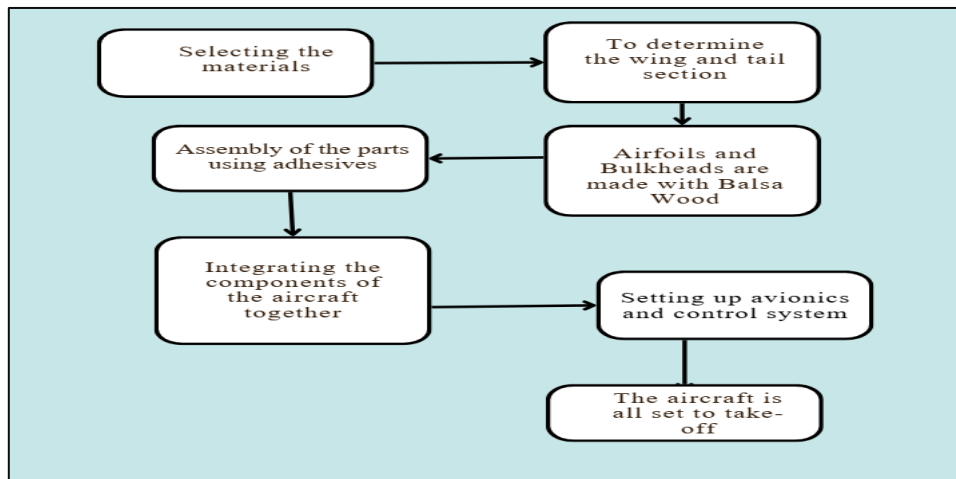


Figure 30 : Methodology of Fabrication

12.1 Material Selection

BALSA WOOD: Balsa Wood has been used for the construction of the ribs and the bulkheads of the wing and fuselage respectively as it has an excellent strength to weight ratio.

ALUMINIUM RODS: The aluminium rods were used for the spars in the wing and also for the fuselage rod because these are very light in weight, and are easily available.

12.2 Fabrication Techniques

LASER CUTTING: The laser cutting technique has been used to obtain the shape of the ribs and bulkheads for the wings and the fuselage.

13 CONCLUSION

The VIHANG aircraft's design and development successfully produced a lightweight, stable, and mission-capable fixed-wing RC platform that satisfies the requirements of the competition. The aircraft provides good lift, stable flying traits, and effective payload operation through careful size, structure, and aerodynamic research, along with ideal powertrain integration. In general, the project increased technical training used in aeromodelling according to the goals of the National Aeromodelling Competition.

Fabrication Photos Zip File Link

https://drive.google.com/drive/folders/1Gvu9hy3H129Hk2z7CHTbgGW900EX_qL6?usp=sharing